PROJECT PRESENTATION ON

PolicyGPT: Government Policy & Public Scheme Intelligence Platform

A platform for discovering, understanding, comparing, and managing Indian government policies and public welfare schemes.

Team-4

Infosys Springboard Internship 7.0

Features and Functionality

Citizen Features

- Explore government schemes and policies in one place
- Search and filter schemes by category, sector, or benefits
- View detailed scheme information and application process
- Check eligibility using demographic and income details
- Compare multiple schemes side-by-side
- Save schemes for future reference
- Receive notifications on new and updated schemes
- Submit feedback on platform and content

Administrator Features

- Create, update, and retire schemes and policies within the platform
- Manage registered user accounts and access permissions
- Monitor platform usage and engagement through analytics dashboards
- Generate periodic and ad-hoc reports for decision-makers
- Configure and manage system-wide notifications and alerts
- Review and respond to user feedback to improve platform quality

Volunteer Features

- Review newly submitted or updated scheme and policy content
- Approve or reject content prior to publication on the platform
- Suggest edits and improvements to ensure clarity and correctness
- Participate in the content verification workflow to maintain data integrity
- Collaborate with administrators to resolve discrepancies and refine information

Technology Stack: MERN

01

Frontend Technologies

- React.js for component-based UI
- Vite for fast, modern builds
- Tailwind CSS for utility-first, responsive styling
- React Router for client-side routing
- Axios for API integration
- Recharts for data visualization

02

Backend & Database

- Node.js for scalable server-side runtime
- Express.js for RESTful backend APIs
- MongoDB as a flexible NoSQL document store
- Mongoose for schema design and data validation

03

Authentication & Communication

- JWT for stateless token authentication
- Bcrypt.js for secure password hashing
- Nodemailer for automated email alerts

04

AI & Data Processing

- AI-powered analysis and insights with Google Gemini
- Reliable, up-to-date policy data via Google Search Grounding
- AI-assisted enrichment of scheme and policy datasets

Outcomes and Impact

Citizen-Centric Benefits

- Centralized access to government schemes and policies in one unified platform
- Simplified discovery of schemes relevant to individual needs and circumstances
- Personalized eligibility assistance through profile-based evaluation and guidance
- Enhanced decision-making by enabling side-by-side comparison of multiple schemes

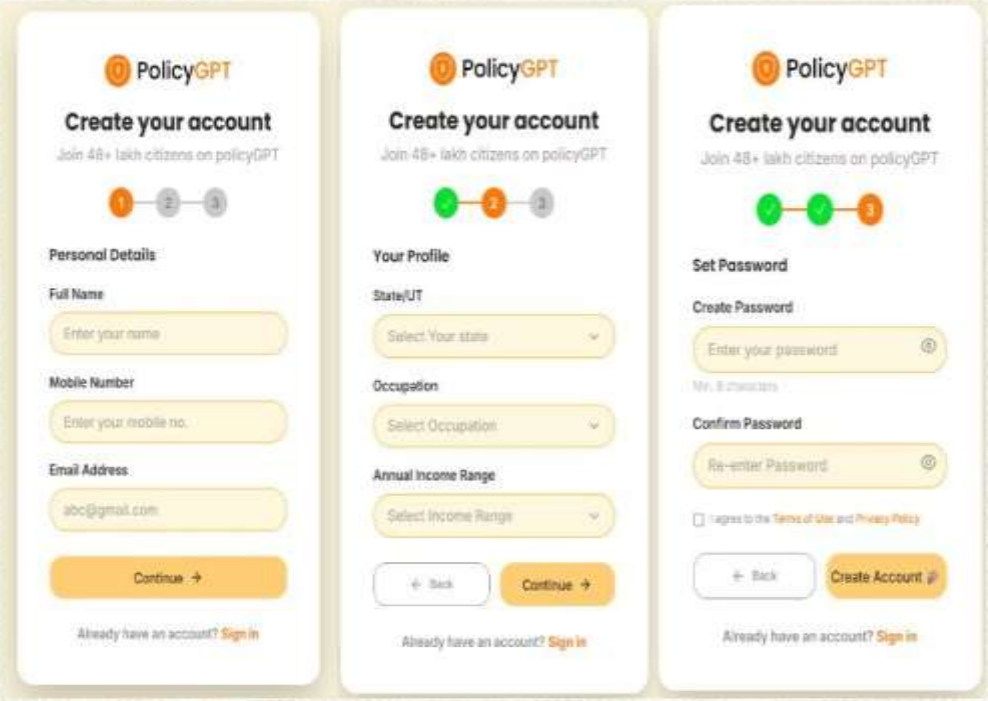
Administrative and Data Impact

- More efficient administration via dedicated tools for scheme, policy, user, analytics, and report management
- Structured content verification enabled by a volunteer and reviewer workflow for information review
- AI-assisted data enrichment using Gemini with Google Search grounding to complete and refine scheme information

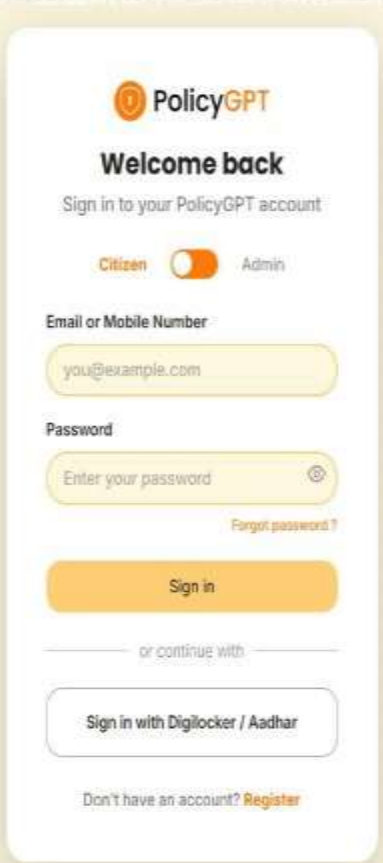
User Roles and Authentication

- JWT-based authentication for stateless, scalable sessions
- Password hashing implemented with Bcrypt.js for secure storage
- Role-based access control enforcing differentiated permissions
- Protected application routes to prevent unauthorized access
- Supports secure registration and login experience

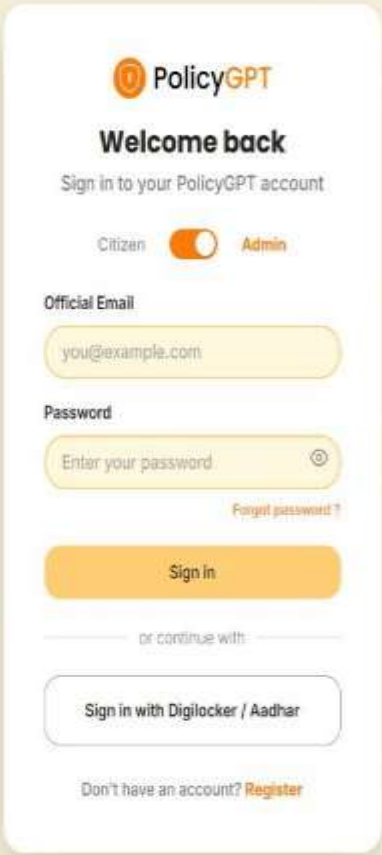
User Registration



Citizen Login



Admin Login



Addressing Challenges

Centralizing Scattered Policy Information

Government schemes scattered across many sites and documents
Citizens and officials lack a clear, comprehensive overview

Solution:
Single, centralized platform for all policy and scheme information
Unified search and navigation to quickly find relevant programs

Simplifying Complex Eligibility Requirements

Eligibility criteria are lengthy, legalistic, and hard to interpret
Citizens can't easily tell if they qualify for specific schemes

Solution: a step-by-step, questionnaire-based eligibility checker
Dynamic questions turn legal rules into clear, user-friendly language

Enriching and Completing Scheme Data

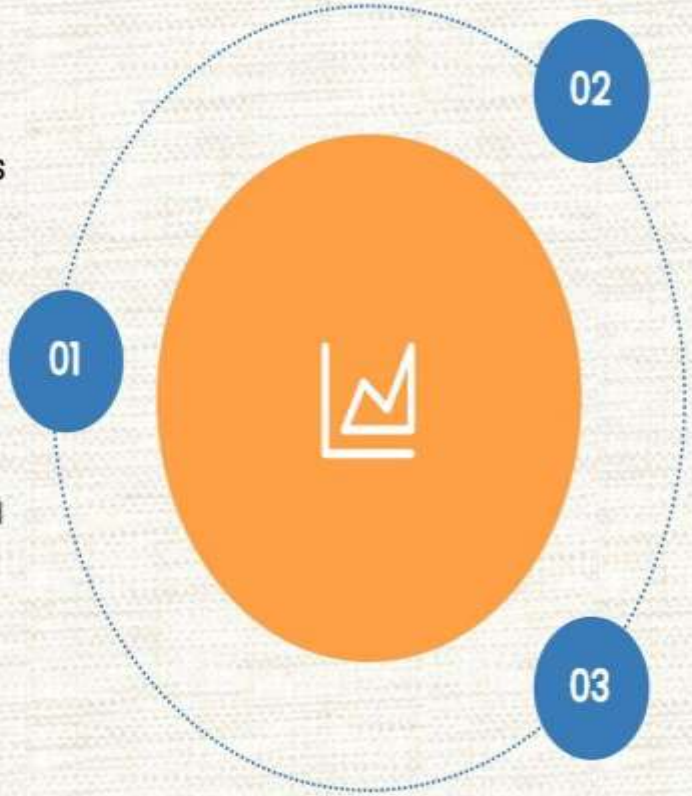
Official scheme descriptions lack detail and clear actions
Missing data blocks comparison and informed decisions

Solution: Gemini-powered data enrichment with Google Search
AI consolidates scattered details into structured, comprehensive scheme profiles

Key Challenges in Platform Design

Multi-stakeholder experience & workflows

- Design distinct yet coherent journeys for citizens, administrators, and volunteers
- Balance citizen simplicity with powerful admin and support tools
- Orchestrate end-to-end scheme and policy approval workflows across roles
- Define clear responsibilities, permissions, and audit trails for each stakeholder



Technical integration & intelligent processing

- Seamlessly connect dashboards to backend REST APIs for real-time data
- Enable dynamic eligibility checks based on policy rules and user data
- Maintain structured, current government policy and scheme information
- Use AI-driven data enrichment to improve completeness, accuracy, and discoverability

UI engineering & collaborative development

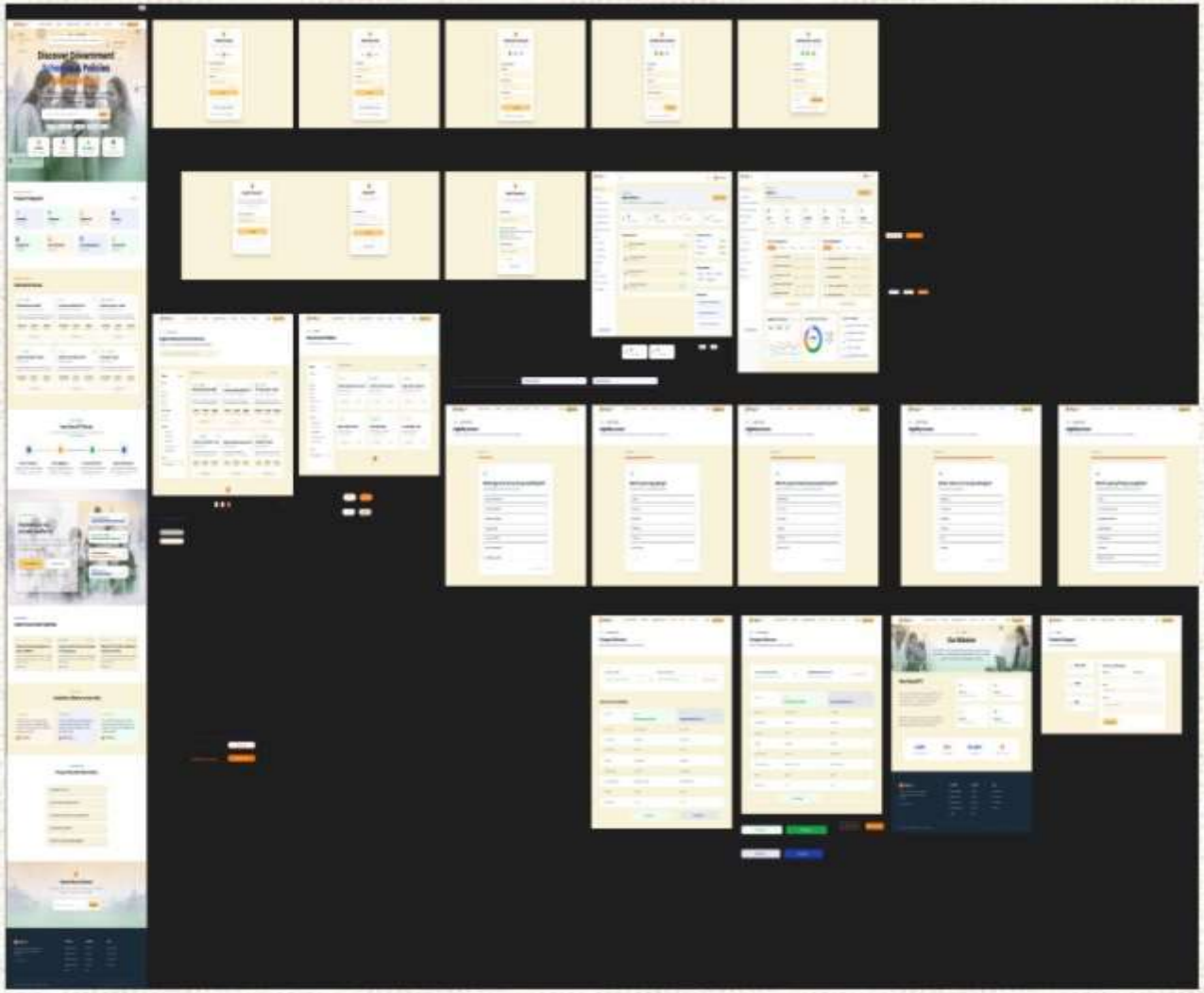
- Build responsive, reusable UI components for all devices and user segments
- Establish a scalable design system to keep interfaces consistent as features grow
- Manage collaborative development with Git/GitHub: branching, reviews, version control
- Coordinate multi-developer contributions while protecting code quality and platform stability

Challenges & Solutions

The table summarizes the key architectural and operational challenges faced in building the PolicyGPT platform and the corresponding technical or process solutions applied to address each one.

Challenge	Solution
Scattered government information	Centralized PolicyGPT platform
Complex eligibility conditions	Automated eligibility checker
Incomplete scheme information	Gemini + Google Search enrichment
Content verification	Volunteer review workflow
Multiple user roles	JWT + role-based access
Large administrative data	Analytics and reporting dashboards
Frontend/backend integration	REST APIs + Axios
Team collaboration	Git + GitHub + Pull Requests

UI/UX Design Approach



Figma-Driven Design Workflow

- Figma for low- and high-fidelity wireframes
- UI layouts planned and refined in Figma
- Dashboards iterated with interactive prototypes
- Component libraries built for scalable, reusable UI
- Clickable prototypes used to validate user flows
- Design handoff streamlined with Figma specs and comments

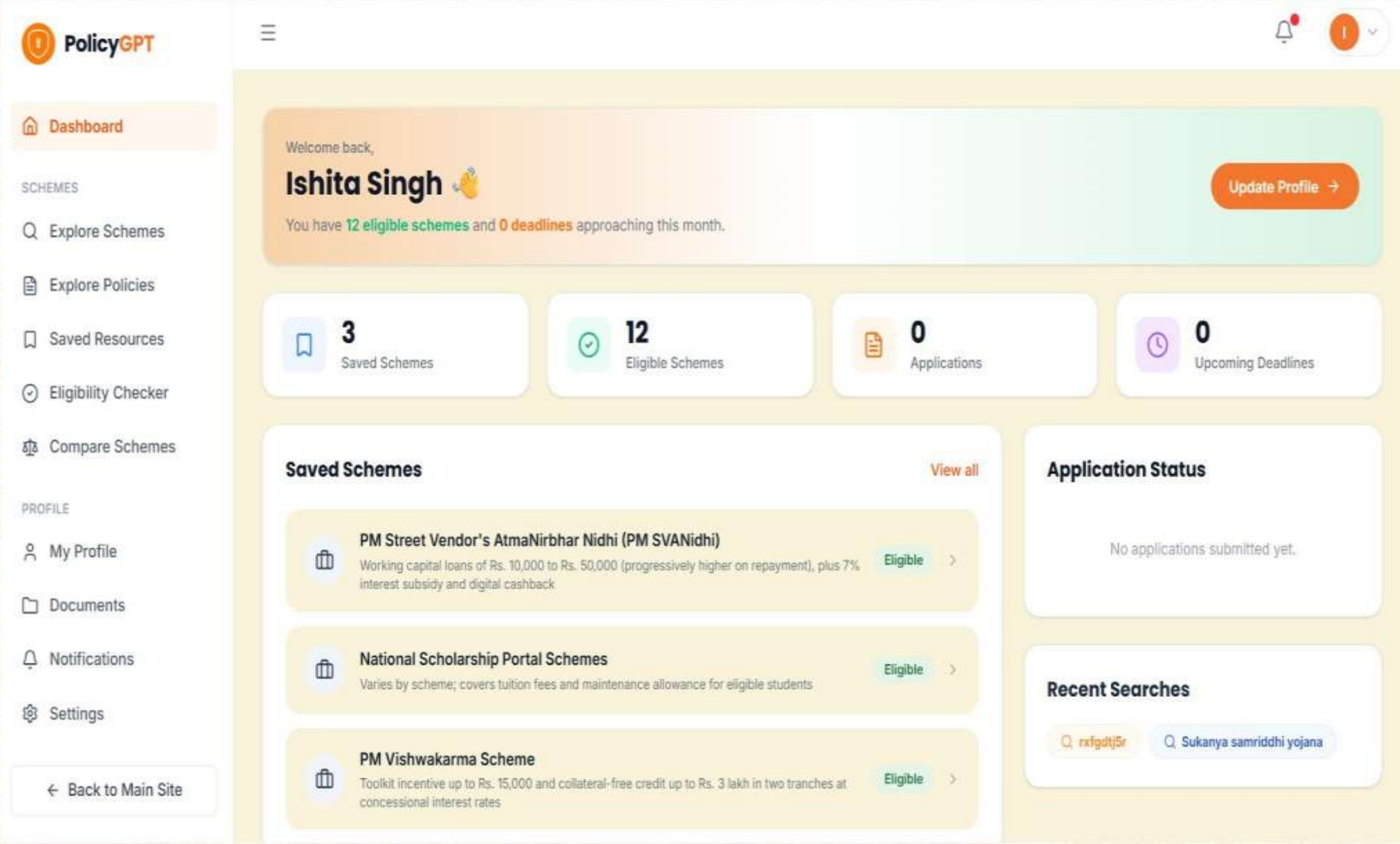
User-Centered Design Focus

- Clean, accessible, user-focused interface for non-technical users
- Simplifies complex government policies and public schemes
- Prioritizes clarity, reducing cognitive load when browsing
- Visual layout enables quick scanning and comparison of policies

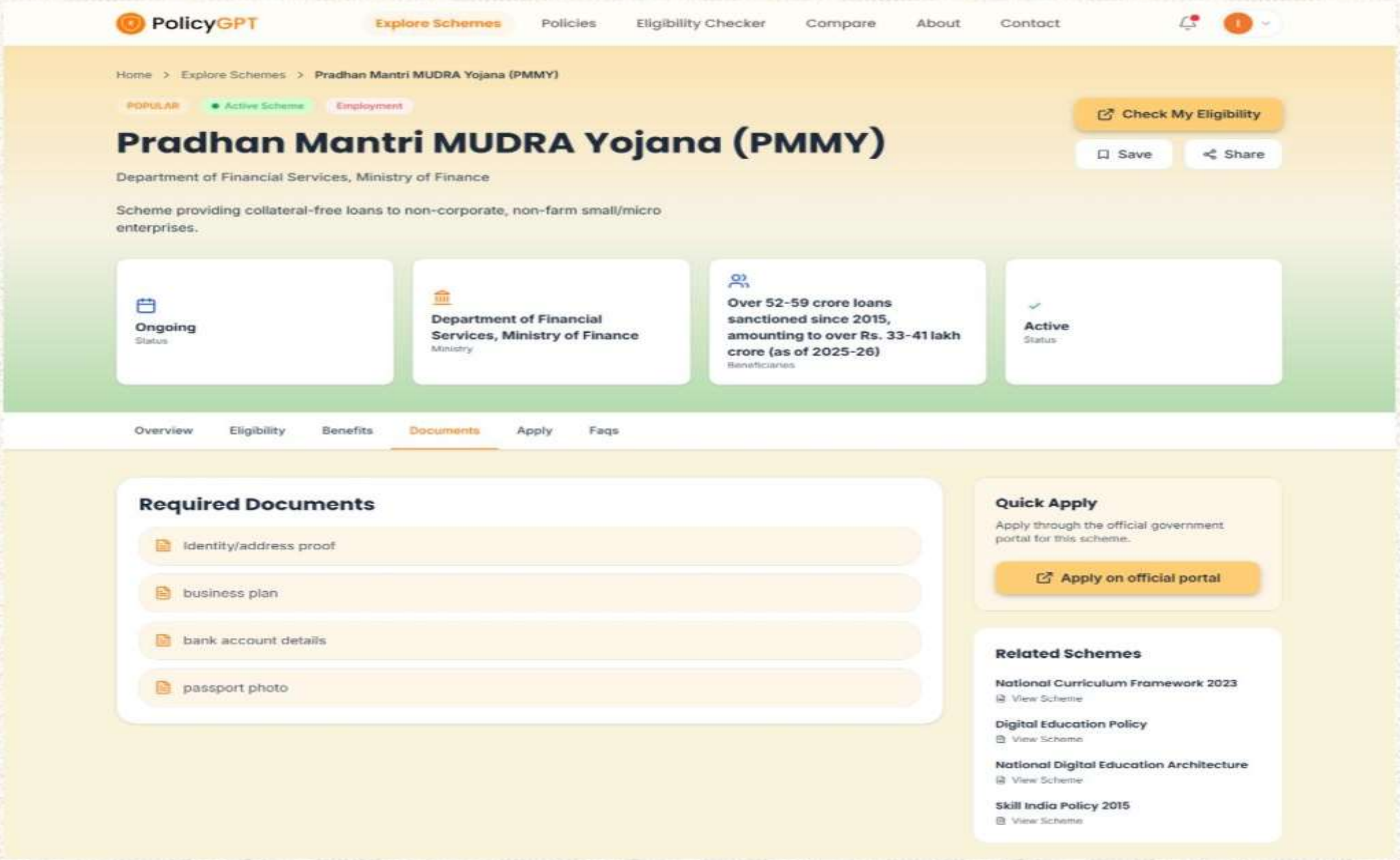
UI Mock-ups: Landing Page



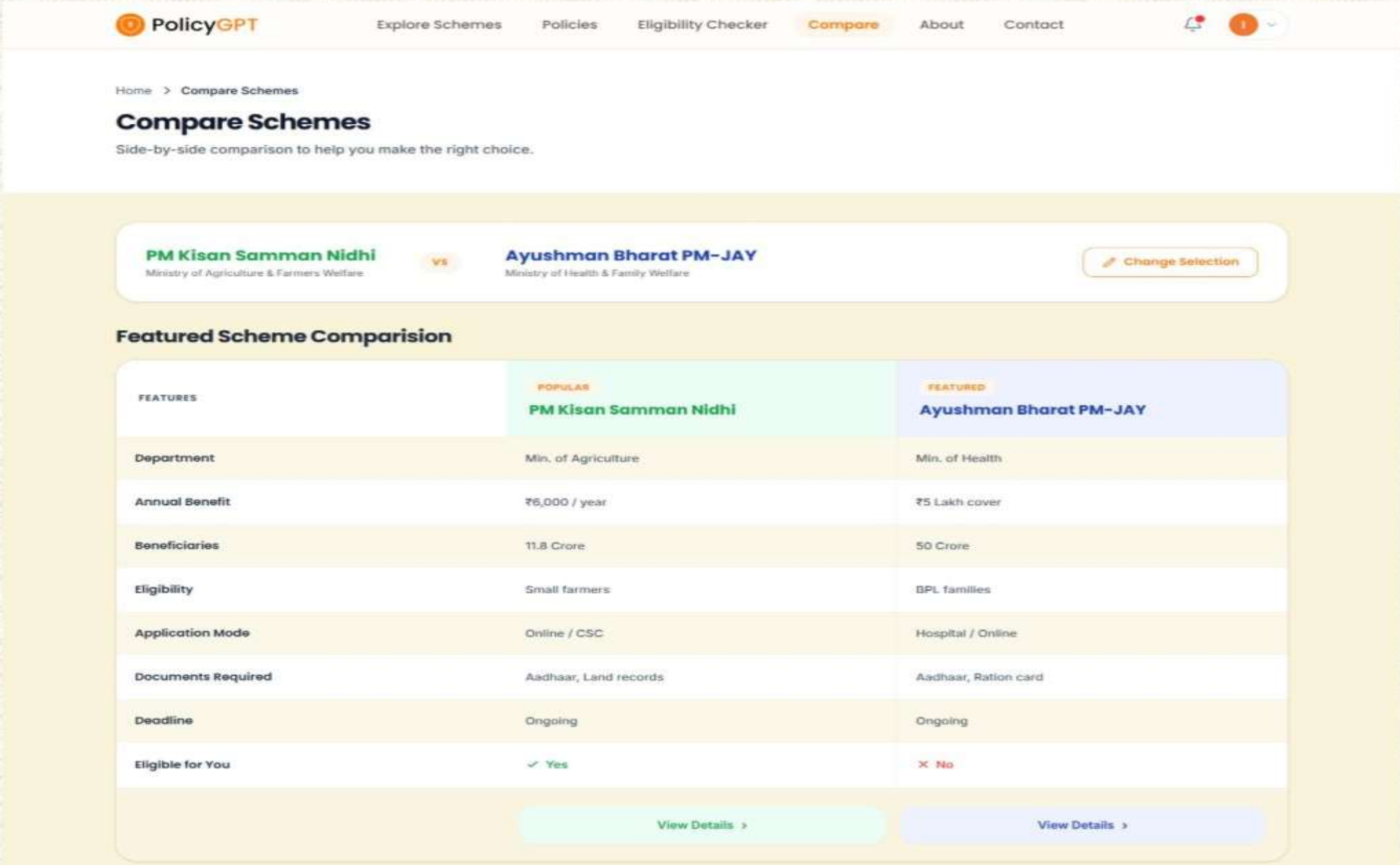
UI Mock-ups: Citizen Dashboard



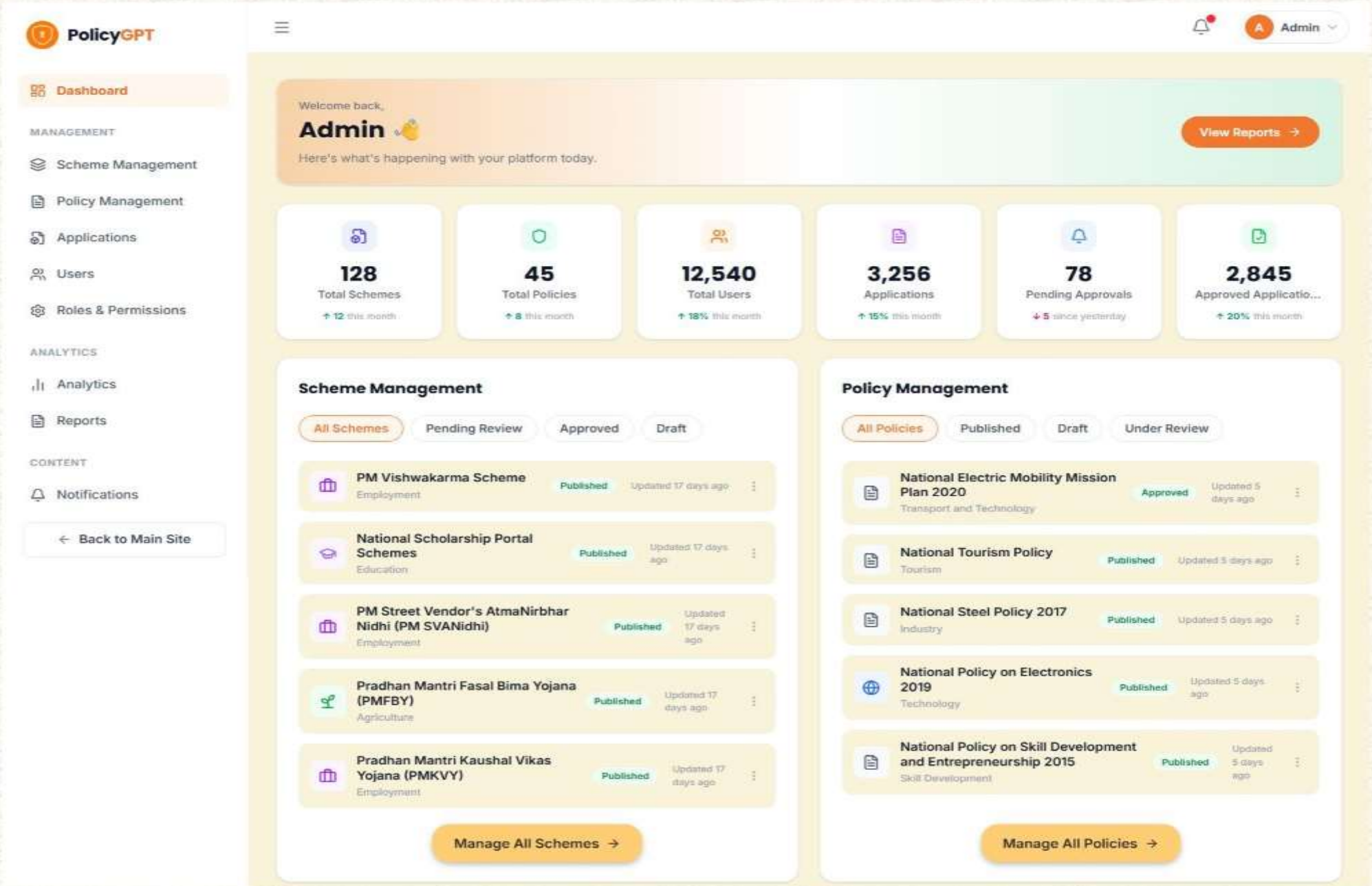
UI Mock-ups: Scheme Details



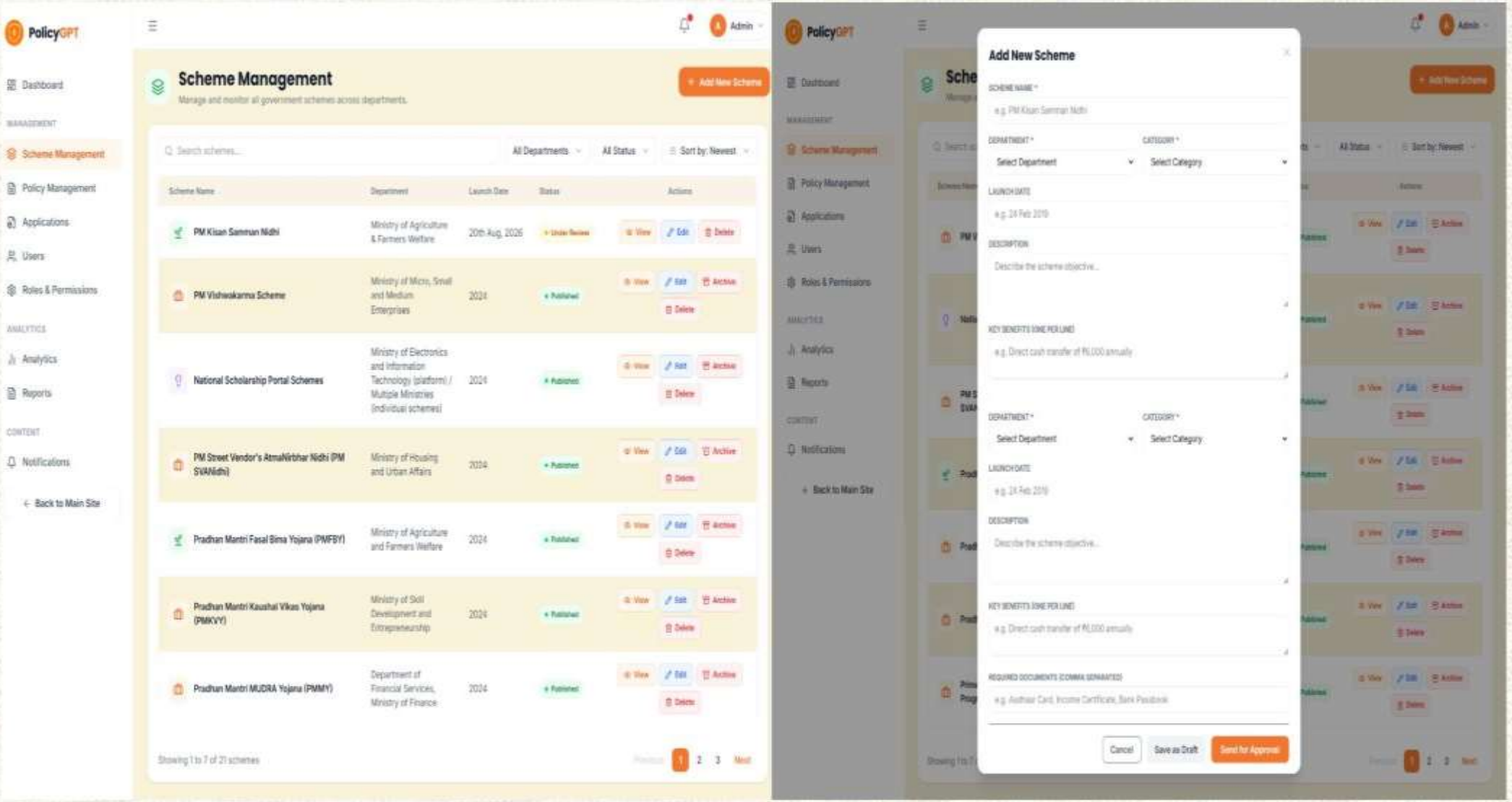
UI Mock-ups: Compare Schemes



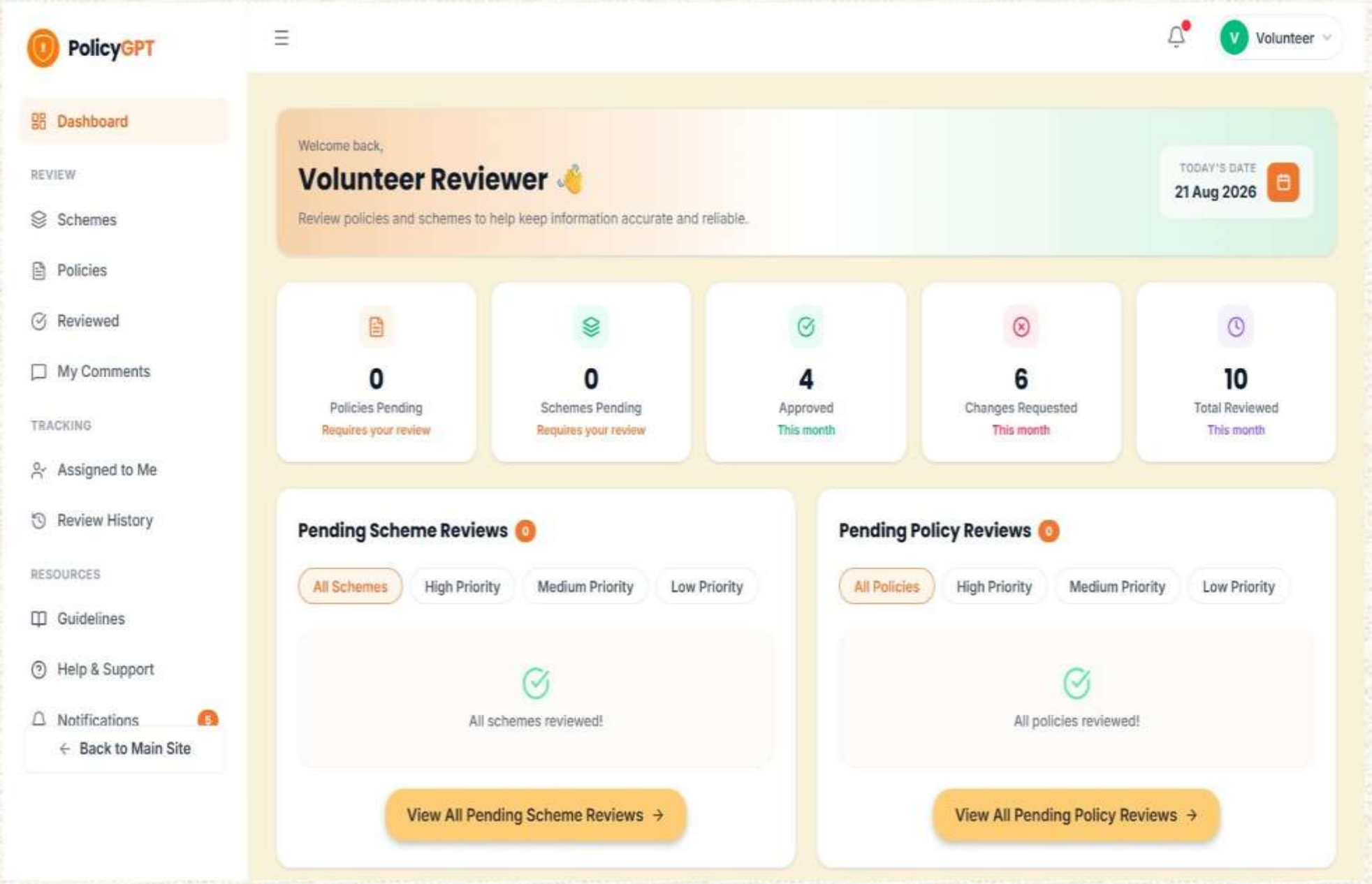
UI Mock-ups: Admin Dashboard



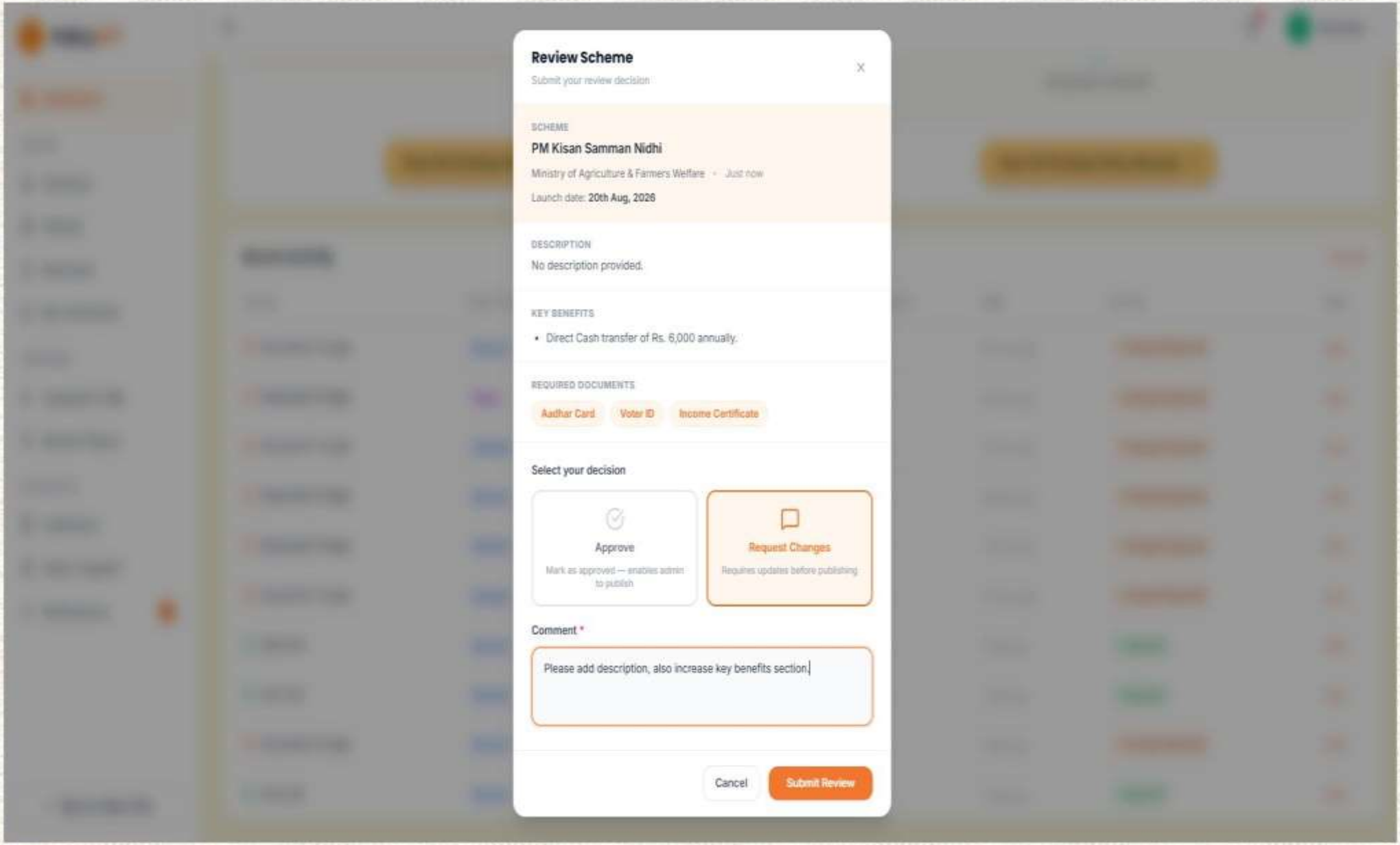
UI Mock-ups: Scheme Management



UI Mock-ups: Volunteer Dashboard



UI Mock-ups: Review Schemes



Our Team



Ishita Singh



Sheetal Gourh



Souvik Chandra



Ankesh Agarwal



Prachi Kumari



Sai Mounika Chunchu



Mythreyee S



Srivadivambigai Sankar

Q&A Session

Thank You!